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Multi-choice best-worst multi-criteria decision-making method and its applications

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Abstract

The best-worst method (BWM) is a multi-criteria decision-making (MCDM) approach for solving several types of real-life decision-making problems. The basic BWM determines the priority order by pairwise comparisons of the best and worst criteria. The strength of the preference assigned via numerical value for linguistic interpretations of relative comparisons ranges from 1 to 9, representing preferences among the criterion. However, the amount of strength is entirely dependent on the choice of the decision-makers. Though expert's opinions may differ due to various reasons such as incomplete information, lack of knowledge, ambiguity in linguistic terms, and so forth. Therefore, it is highly likely that the expert may provide multiple viewpoints for the preferences values. Thus, this study proposes a novel extension of the BWM named the multi-choice best-worst method (MCBWM), dealing with the concept of multiple choices of preference relations to compare the criterion. The MCBWM overcomes the limitation of BWM, where the pairwise preferences in the comparisons are multi-choice parameters rather than single parametric values. Multiple real-world MCDM applications are illustrated in experimental studies to show the quality, performance, and applicability of our proposed MCBWM. A detailed comparative analysis of the proposed approach has been done with the well-known existing decision-making techniques and their optimal results are compared. The proposed method offers a new direction for the MCDM approaches for solving real-life problems.

KEYWORDS

best-worst method, cloud server selection, multi-choice best-worst method, multi-choice programming approach, multi-criteria decision-making

1 | INTRODUCTION

Multi-criteria decision-making (MCDM) plays a significant role in the decision-making of real-life problems. MCDM approaches determine the ranking of alternatives and chooses the superior one using a suitable approach based on some criteria.[1] It has a wide range of applications in multiple fields such as engineering, economics, social sciences, and management.[2–4] Generally, the MCDM procedure consists of the following steps: (a) formulation of the decision-making problem, (b) defining a set of criteria, (c) constructing a decision metric, (d) calculation of criteria weight, and (e) ranking of alternatives.

Many MCDM methods are prevailing in literature, such as data envelopment analysis (DEA),[5,6] analytic hierarchy process (AHP),[7,8] elimination and choice expressing reality (ELECTRE),[9,10] VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR),[11,12] preference ranking organization method for enrichment evaluations (PROMETHEE),[13,14] decision-making trial and evaluation laboratory (DEMATEL), technique for order of preference by similarity to ideal solution (TOPSIS),[13,15] analytic network process (ANP)[16,17] and best-worst method (BWM).[18,19] Though there are several approaches to handle an MCDM problem, the best suitable one must depend upon the structure of the decision-making problem.

Most of the MCDM methods consider exact and crisp values of input data. But It's not always possible to gather exact input data, and in such cases, uncertainty arises. To deal with uncertainty, many extensions of MCDM methods were found in the literature. There are three major types of uncertainty that may occur in input data. It may arise due to ambiguity, stochasticity, and due to partial information. To handle such uncertainties in input data, MCDM methods are extended using specific techniques such as fuzzy sets, Intuitionistic fuzzy sets, Hesitant fuzzy sets, Interval-valued fuzzy sets, Interval-valued Hesitant fuzzy sets, Interval-valued Intuitionistic fuzzy sets, type-2 fuzzy sets, Interval type-2 fuzzy sets, Linguistic variables, Evidential reasoning, Rough sets, and Probability theory, and so forth. Some of the work prevalent in literature with different types of input data are Proportional interval type-2 hesitant fuzzy,[20] hesitant fuzzy linguistic term set,[21] D-number,[22,23] Z-number,[24] Dempster–Shafer theory,[25,26] intuitionistic fuzzy,[27,28] neutrosophic,[29,30] and probability theory,[31,32] and so forth.

Rezai[18] introduced and developed an MCDM technique named the BWM. BWM determines the pairwise relative comparisons, that is, the preference between the best criterion over all other criteria and the preference between all criteria over the worst criterion. On comparing

AHP[7] with the BWM approach, BWM utilizes fewer calculations with lesser pairwise comparisons than the AHP. As a result, BWM presents a low inconsistency of pairwise comparisons against AHP, hence a more reliable method. All the pairwise comparisons are obtained by assigning a number through a scale ranging from 1 to 9. Since the trivial comparisons are not performed, the BWM seems more convenient, precise, and far less redundant.[18] A typical illustration of the reference comparison is shown in Figure 1.

Suppose $A = o_{ij}$; $i, j = 1, 2, 3, ..., n$ be a pairwise comparison matrix to compare the importance of n criteria $C_1, C_2, C_3, ..., C_n$ and constructed based on the pairwise comparison using[18] as follows:

$$A = \begin{array}{c} \\ C_1 \\ C_2 \\ \vdots \\ C_n \end{array} \begin{array}{c} \begin{array}{cccc} C_1 & C_2 & \cdots & C_n \end{array} \\ \begin{pmatrix} o_{11} & o_{12} & \cdots & o_{1n} \\ o_{21} & o_{22} & \cdots & o_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ o_{n1} & o_{n2} & \cdots & o_{nn} \end{pmatrix} \end{array} \tag{1}$$

where o_{ij} represents the degree of relative importance (or preference) of criterion C_i over criterion C_j, the values of o_{ij} are range between the 1/9 to 9 scale. For an equal preference of criterion C_i over C_j, o_{ij} will take value 1. $o_{ij} \geq 1$ shows that criterion C_i is relative more importance than criterion C_j with $o_{ij} = 9$ exhibiting the extreme preference of criterion C_i over C_j. The relative importance of C_j over C_i is given by o_{ji}. For matrix A to be reciprocal, the relative importance of C_i over C_i is one ($o_{ii} = 1$) and $o_{ij} = 1/o_{ji}, \forall i, j$. For A to be perfectly consistent $o_{ik} \times o_{kj} = o_{ij}, \forall i, j$.

In pairwise reference comparisons (see Figure 1), both direction and strength of the preference of criterion C_i over criterion C_j are represented by the decision-maker (DM). In most circumstances, there is no problem in conveying the direction by the DM. However, representing the strength of the preferred choice is a difficult task, which is the primary source of inconsistency in the pairwise comparison matrix (A). Since the level of preference relation is assigned with a numeric number, it may be possible that the provided value is insufficient to accommodate the uncertainty with the linguistic terms. Moreover, while taking preference from DMs, a difficulty may occur in the information in terms of choices. In other words, it is possible that different DM gives different importance for the desired reference comparisons to compare particular criteria with the others. DM may provide multiple responses for the pairwise comparison of criterion C_i over C_j. Thus, based on this, considering the pairwise reference

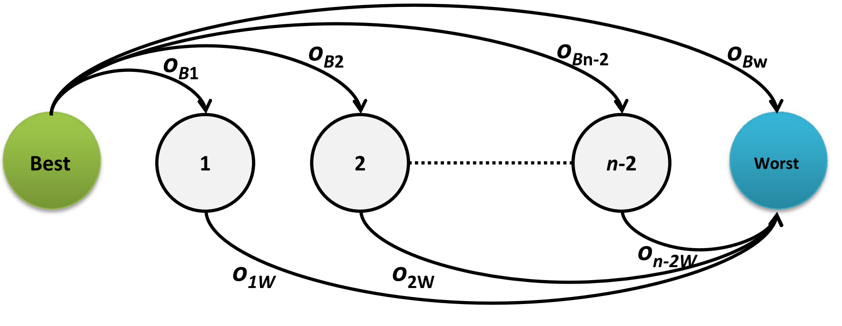

FIGURE 1 Pairwise reference comparisons[18] [Color figure can be viewed at wileyonlinelibrary.com]

comparisons as multi-choice comparisons is highly reasonable. This approach benefits with looking into the consistency in the judgments of the importance of the decision. The multiple choices for criteria lead to accommodate all possible opinions regarding the pairwise comparison of a particular criterion by the DM.

This paper introduces a new MCDM approach named the multi-choice best-worst method (MCBWM) using the multi-choice comparisons. The multi-choice comparison became crucial in the MCDM problem when a DM is uncertain about the importance of a particular criterion over another. Multi-choice parameters in the reference comparison will improve flexibility in decision-making. MCBWM provides more relaxation to the DMs to use more than one pairwise comparison due to lack of information, confusing features of criteria, environment conditions, and perspective to make the judgment. Further, the proposed approach decreases the inconsistency that provides support to the DMs for clearly understanding the accuracy and correctness regarding the final ordering.

The significant contributions of the work are composed as follows:

1. It proposes a novel MCBWM based on multi-choice pairwise reference comparisons for more realistic decision-making problems.
2. The proposed MCBWM technique suitably enhances decision-makers importance as they are open to provide more than one degree of preference to compare a criterion.
3. To obtain the optimal weights of the criteria through the optimization model, an efficient multi-choice programming approach and Lagrange interpolation polynomial are introduced.
4. To evaluate the effectiveness of the introduced MCBWM procedure, three case studies of transportation of goods, selecting the high-performance car, and cloud service selection are evaluated considering multiple choices in optimization models.
5. The best criteria among multiple combination models are then used to precisely determine the ranking, whose values achieve the minimum consistency ratio.
6. The experimental simulations present the proposed MCBWM's strengths and weaknesses, which capably decreases the inconsistency in building a robust decision-making model.

The remaining paper organization is as follows: a comprehensive survey of the BWM and other MCDM methods is completed in Section 2. Section 3 presents the proposed MCBWM method. In Section 4, experimental simulation is conducted using the proposed approach, and the results are compared with the existing ones. Finally, we conclude in Section 5 and discuss the future directions of the work.

2 | LITERATURE SURVEY

The BWM is a newly developed MCDM technique proposed by Rezaei[18] in 2015. The author[18] has applied the BWM scheme for a mobile phone selection problem. Following its inception, numerous researchers successfully used BWM methodology to solve many real-world decision-making problems. Rezaei et al.[33] proposed linking supplier segmentation to enhance the logical supplier development model where the segmented suppliers are incorporated by the BWM approach. A BWM is introduced for finding the most significant enablers of technological innovation in the sense of Indian micro-small, and medium-sized enterprises (MSMEs) by Gupta and Barua.[34] A supplier selection problem related to the food supply chain of the edible oils industry, Rezaei et al.[35] used the BWM to identify the best suppliers. Rezaei et al.[35] utilized the BWM in an MCDM problem for complex bundling configurations in the surface transportation of air freight. Ghaffari et al.[36] analyzed the critical success factors in the development of remotely-piloted helicopters in Iran using BWM. Groenendijk et al.[37] applied the BWM to obtain the weight of criteria in the decision-making manner of discovering the most suitable approach to improve public transportation quality in a case study in Rotterdam. Sadaghiani et al.[38] investigated the importance of external forces on the supply chain sustainability in the oil and gas industry through the BWM.

Further, because of its high consistency and the limited amount of pairwise comparisons required, BWM is applied in different real-world cases such as failure mode and effects analysis,[39,40] decision-making in the judgment of scientific journal and investment projects,[41] water scarcity management in arid regions,[42] evaluation and classification of overseas talents in China,[43] multi-criteria sustainability assessment of technologies urban sewage sludge,[44] assessing the social sustainability of supply chains management,[45] risk assessment framework for business continuity management systems,[46] decision-making in web service selection,[47] selection of the mobile phone,[18,32] and determining the location of facilities.[48,49] Enthusiastic readers can refer to [50] to study more about the current advancements in the BWM and its applications. Mi et al.[50] provided a detailed state-of-the-art on BWM since Rezaei introduced it. Efforts in enhancing the BWM to achieve a worthwhile modification for the real-world problems are still going on. The primary BWM is proposed as a nonlinear optimization problem[18] which approximated as a linear framework[19] and further extended in multiplicative framework,[51] interval version,[52] Bayesian framework,[32] and so on. Several hybrid versions are also proposed in the literature to enhance the performance MCDM using BWM such as BWM-TOPSIS,[53] BWM-MULTIMOORA,[52] BWM-VIKOR,[54] BWM-ELECTRE,[55] BWM-FDA,[56] and so forth. Zhang et al.[57] propose a cognitive-based BWM, where they have used an interval scale for pairwise comparisons. It combines two approaches, Primitive Cognitive Network Process and BWM method.

In literature, many researchers extended the BWM from different theoretical aspects to solve various decision-making problems under unusual circumstances. Since the inherent ambiguity in decision-making procedures plays an important role, the potential of fuzzy theory to incorporate uncertain information and flexibility in human judgment has recently been investigated in decision-making using BWM under the imprecise environment. In this context, a fuzzy BWM was proposed by Guo and Zhao,[58] in which fuzzy number represents pairwise reference comparisons with linguistic terms. To manage the uncertain pieces of information associated with an MCDM, Aboutorab et al.[24] extended the BWM by incorporating Z-numbers. Mou et al.[27] suggested intuitionistic fuzzy multiplicative BWM to determine the weight's criteria through a nonlinear mathematical model for health care management problems. An intuitionistic fuzzy multiplicative preference relation is used for comparing the pairwise comparisons. Hafezalkotob and Hafezalkotob[41] proposed a fuzzy BWM for group decision-making in a company with a superior decision-maker that maintained a group of experts. Recently, In another application of group-BWM, Ahmed et al.[59] applied the approach for identification and prioritization of strategies to handle the Covid-19 outbreak. Pamučar et al.[60] utilized interval-valued fuzzy-rough numbers to develop modified multi-attributive border approximation area comparison and applied it to the case study determining the optimal model of fire-fighting helicopters.

Karimi et al.[61] studied repair assessment at a town hospital in northeastern Iran using a linear mathematical model by implementing a fully fuzzy BWM process. An evidential BWM is proposed by Fei et al.[62] to calculate the weight of criteria by applying the theory of belief function for a hospital service evaluation problem. Wu et al.[63] proposed an interval type-2 fuzzy BWM and develop optimization models to get the weights of criteria and applied it to the green supplier selection problem. To shortlist, rank, and select the best cloud service, Hussain et al.[64] developed an integrated methodology using BWM in an optimal service selection (MOSS) in a cloud environment. The same authors in [65] proposed a fuzzy linear BWM framework for service selection for Cloud Service Selection as a Service. Ahmad and Hasan[66] have utilized the fuzzy BWM approach to prioritize and find weights of mobility, node density, energy, and node status for countering the destination unreachability problem. In another type of BWM, best to others and others to worst preferences are collected along with the respective confidence level as single-valued trapezoidal neutrosophic number.[67] Chen et al.[68] proposed a hybrid BWM method under a rough fuzzy environment. A belief-based BWM considering the level of belief of preferences is proposed by Liang et al.[69] Sen et al.[70] applied the combination of BWM and Dempster–Shafer theory for a flood resilience development model for housing infrastructure. A risk-based BWM based on R-numbers is applied in a project selection problem in a healthcare firm.[71] A BWM integrated to an interval rough numbers have been applied to a waste disposable location problem.[72]

Based on the above discussion, we see that the bibliography of the BWM is very rich and has so much scope to work. Researchers have not been considered a most critical point that is multiple opinions as choices in preference comparisons.

As mentioned earlier, the issue is vital in the decision-making of real-world and large-scale problems and is computationally costly. Therefore, the present work focuses on a multi-choice mathematical programming model to counterbalance the criteria and alternatives. The subsequent section illustrates the proposed methodology.

3 | PROPOSED APPROACH

Similar to other MCDM methods, there are different variants of BWM under uncertain environments. Several extensions of the BWM under fuzzy, Hesitant fuzzy, Type-2 fuzzy, Bayesian, Markov chain, Rough set theory, Dampster–Shafer theory, D-number, Z-number, R-number, and many more uncertain environments were proposed in prevalent literature. The fuzzy BWM[58] is based on triangular fuzzy number $\tilde{a}_i = (l_i, m_i, u_i)$ as preference comparison, which is further ranked using graded mean integration representation $R_{a_i} = \frac{l_i + 4m_i + u_i}{6}$. Fei et al.[62] proposed an evidential BWM based on belief function theory having evidential reference comparisons. A belief degree is employed to express the confidence level that each criterion or combination of criteria is the best one or worst one. They have shown an application in hospital service quality evaluation problems. Liao et al.[73] proposed a hesitant fuzzy linguistic term-based BWM. Recently, Liu et al.[74] proposed D-numbers-based BWM approach. They have considered a D number extended fuzzy preference relations for pairwise comparison of criteria. Zhang et al.[75] combine the BWM with Dempster–Shafer theory considering a newly developed Double hierarchy hesitant fuzzy linguistic term set model. The extensions of BWM using fuzzy set theory, type-2 fuzzy, hesitant fuzzy, intuitionistic fuzzy, and rough-fuzzy theory are complicated and may lose information due to the involvement of defuzzification. Also, as the level of uncertainty increases, the complexity of the solution process increases. All extensions of BWM under uncertain environments have their own relevance, importance, and applications. They are applied when the assumptions of BWM-based approach in a particular problem are satisfied.

Keeping all this in mind, we have developed a MCBWM under the assumption that single choice or multiple choices are provided by the decision-maker for the reference comparison of criteria. In literature, as per our knowledge, there is no such work that considers the preference comparison between the criteria as a set of multiple choices without putting restriction of having three and four choices like in triangular and trapezoidal fuzzy number, respectively.[58] Also, no work considers multiple choices in reference comparisons without any belief degree of reference comparison between the criteria like in evidential BWM.[62] The objective of BWM is to minimize the inconsistency of the information obtained from the experts. The concept of proposed multi-choice BWM considers those choices from the sets of choices of pairwise comparisons, for which the inconsistency is minimized. In the next sub-sections, the formulation of multi-choice programming, BWM, and finally proposed MCBWM is presented.

3.1 | Multi-choice programming problem

A MCPP is a type of mathematical programming problem where the requirement is to choose an alternative from several possible combinations of parameters to optimize an objective function subject to a set of constraints.[76–78] The situation where a parameter consists of multiple choices exists in many managerial real-life decision-making problems. It helps to decide the best collection of parameter values from various parameter values. Mathematically, a MCPP[79,80] can be represented as:

$$\begin{aligned} &\text{Min / Max:} \quad \sum_{j=1}^{n} \left\{c_j^1, c_j^2, \ldots, c_j^{k_j}\right\} x_j \\ &\text{subject to} \\ &\sum_{j=1}^{n} \left\{a_{ij}^1, a_{ij}^2, \ldots, a_{ij}^{p_{ij}}\right\} x_j \le \left\{b_i^1, b_i^2, \ldots, b_i^{r_i}\right\}, i = 1, 2, \ldots, m \\ &x_j \ge 0, j = 1, 2, \ldots, n. \end{aligned} \tag{2}$$

In the above formulated model, $c_j^l, l = 1, 2, \ldots, k_j$ is the alternative value of multi-choice cost coefficient $c_j, j = 1, 2, \ldots, n$, $a_{ij}^s, s = 1, 2, \ldots, p_{ij}$ be the multi-choice technological coefficient of $a_{ij}, i = 1, 2, \ldots, m; j = 1, 2, \ldots, n$, and $b_i^t, t = 1, 2, \ldots, r_i$ is the multi-choice resource variable for $b_i, i = 1, 2, \ldots, m$. The decision variables $x_j, (j = 1, 2, \ldots, n)$ are deterministic in the problem. We take the help of interpolating polynomials to tackle the multi-choice parameter in MCPP.

3.2 | BWM

The BWM uses five simple steps to derive the weights of criteria.[18] Initially, the n criteria for an MCDM are determined. The decision-maker (DM) determines the best and worst criterion out of the set of criteria. After that, pairwise comparisons between the best criterion to others and other criteria to the worst are performed, as shown in Figure 1. The importance of pairwise comparisons on those criteria is achieved by the linguistic terms (variables) from the DM, such as equally Important (EI), Moderately Important (MI), Important (I), Extremely Important (EXI), and their intermediate correspondence. Then, the DM asked to convert such linguistic terms into numeric values for evaluations, and the rules of transformation are listed in Table 1. The scale varies from equally important to extremely important, with numeric values from 1 to 9. A mathematical model has been formulated based on these comparisons and solved to obtain the optimal weights of criteria.

We summarize the five step procedure of BWM as follows:

Step 1. Determine the set of decision criteria $\{C_1, C_2, ..., C_n\}$.

Step 2. Select the best criterion (C_B) and the worst criterion (C_W) from the set of criteria.

Step 3. The pairwise comparison between the best criterion (C_B) and all other criteria is performed resulting best-to-others (O_B) vector as shown in Equation (3) as

$$O_B = (o_{B1}, o_{B2}, o_{B3}, ..., o_{Bn}), \tag{3}$$

where, o_{Bj}; $o_{Bj} \geq 1, j = 1, 2, ..., n$ represents the degree of preference regarding the best criterion C_B with other criterion $C_j, j \neq B$.

Step 4. The others-to-worst (O_W) vector presenting the pairwise comparison of preferences between all criteria and the worst criterion (C_W) is given in Equation (4) as

$$O_W = (o_{1W}, o_{2W}, o_{3W}, ..., o_{nW})^T, \tag{4}$$

where, o_{jW}; $o_{jW} \geq 1, j = 1, 2, ..., n, j \neq W$ represents the degree of preference regarding the worst criterion C_W with other criterion C_j.

Step 5. Calculate the weights of criteria by formulating an optimization model. To calculate the weights for ranking of criteria, the optimal weights must satisfy $W_B/W_j = o_{Bj}$ and $W_j/W_W = o_{jW}$. Thus, the objective is to minimize the absolute maximum difference of $|W_B/W_j - o_{Bj}|$ and $|W_j/W_W - o_{jW}|$. Then, a min-max optimization model is formulated as follows:

Model 1:

$$\begin{aligned} &\text{Min}\ \underset{j}{\text{Max}} && \{|W_B/W_j - o_{Bj}|, |W_j/W_W - o_{jW}|\} \\ &\text{subject to} && \sum_j W_j = 1,\ W_j \geq 0,\ \forall j = 1, 2, ..., n. \end{aligned} \tag{5}$$

Here, W_B and W_W are the weights of best and worst criterion, respectively. The weight of criterion C_j is represented by W_j. The linear programming model of model 1 is transformed as

TABLE 1 Linguistic variables for pairwise comparison

Scale	Linguistic term	Scale	Linguistic term
1	Equally Important (EI)	2	Intermediate of EI & MI (IEM)
3	Moderately Important (MI)	4	Intermediate of MI & I (IMI)
5	Important (I)	6	Intermediate of I & VI (IVI)
7	Very Important (VI)	8	Intermediate of VI & EXI (IEI)
9	Extremely Important (EXI)		

Model 2:

$$\begin{aligned} \text{Min} \quad & \xi \\ \text{subject to} \quad & |W_B - W_j \times o_{Bj}| \le \xi \\ & |W_j - W_W \times o_{jW}| \le \xi \\ & \sum_j W_j = 1, W_j \ge 0. \end{aligned} \tag{6}$$

The optimal values of ξ are utilized to determine the consistency ratio. A comparison is said to be fully consistent, when $o_{Bj} \times o_{jW} = o_{BW} \ \ \forall j$.

3.3 | MCBWM

This section presents a better understanding of multi-choice parameters in relative preferences, which is multiple opinions in the comparison. The multi-choice parameters became crucial in the MCDM problem when an expert doubts the importance of a particular criterion. Because of the lack of information, confusing features of criteria, environment conditions, and perspective of MCDM problem cause difficulty to the DM in making the judgment. We extended the BWM for the multi-choice comparisons to introduce the MCBWM. The complete procedure of the proposed MCBWM is presented in Figure 2. Multi-choice parameters in the reference comparison improve the flexibility and provide more relaxation to use more than one choice to compare a criterion over other. The MCBWM utilizes the MCPP approach of pairwise comparison results in less inconsistency.

From Equations (3) and (4), the pairwise reference comparisons, o_{Bj} and o_{jW}; $j = 1, 2, ..., n$ are chosen by the DMs on a scale 1–9 as given in Table 1. However, one will come across with the multiple possibility regarding such choices. Thus, best-to-others (O_B) and others-to-Worst (O_W) vector preferences may take multiple choices from the experts as shown in Figure 3. In other words, each pairwise reference comparison o_{Bj} and o_{jW} for the jth criterion C_j may consist with k number of choices, that is, $o_{Bj} = \{o^1_{Bj}, o^2_{Bj}, o^3_{Bj}, ..., o^k_{Bj}\}$ and $o_{jW} = \{o^1_{jW}, o^2_{jW}, o^3_{jW}, ..., o^k_{jW}\}$, respectively. The reference comparison best-to-others (O_B) and others-to-worst (O_W) vectors under the assumption of multi-choice parameter is defined as $O_B = (\{o^k_{Bj}\}; \forall j = 1, 2, ..., n)$ and $O_W = (\{o^k_{jW}\}; \forall j = 1, 2, ..., n)$, respectively such that $k \subseteq \{1, 2, 3, ..., 9\}$.

Hence, the steps of the proposed MCBWM to obtain the weights of criteria are as follows:

Step 1. Initially, the DM select the most important, that is, the best criterion (C_B) and the least important, that is, the worst criterion (C_W) from the given set of n criteria $\{C_1, C_2, ..., C_n\}$.

Step 2. Perform the pairwise comparison of preferences between the best criterion C_B over all other criteria $C_j; j = 1, 2, ..., n$. Then, best-to-others (O_B) vector having a set of k number of choices is shown in Equation (7) as

$$O_B = (o_{B1}, o_{B2}, o_{B3}, ..., o_{Bn}), \quad \text{where } o_{Bj} = \left\{o^1_{Bj}, o^2_{Bj}, o^3_{Bj}, ..., o^k_{Bj}\right\} \quad \forall j, \tag{7}$$

where, o^k_{Bj} represents the collection degree of multi-choice preference regarding the best criterion C_B with other criterion C_j such that $o^k_{Bj} \ge 1, j = 1, 2, ..., n$ and $k \subseteq \{1, 2, 3, ..., 9\}$.

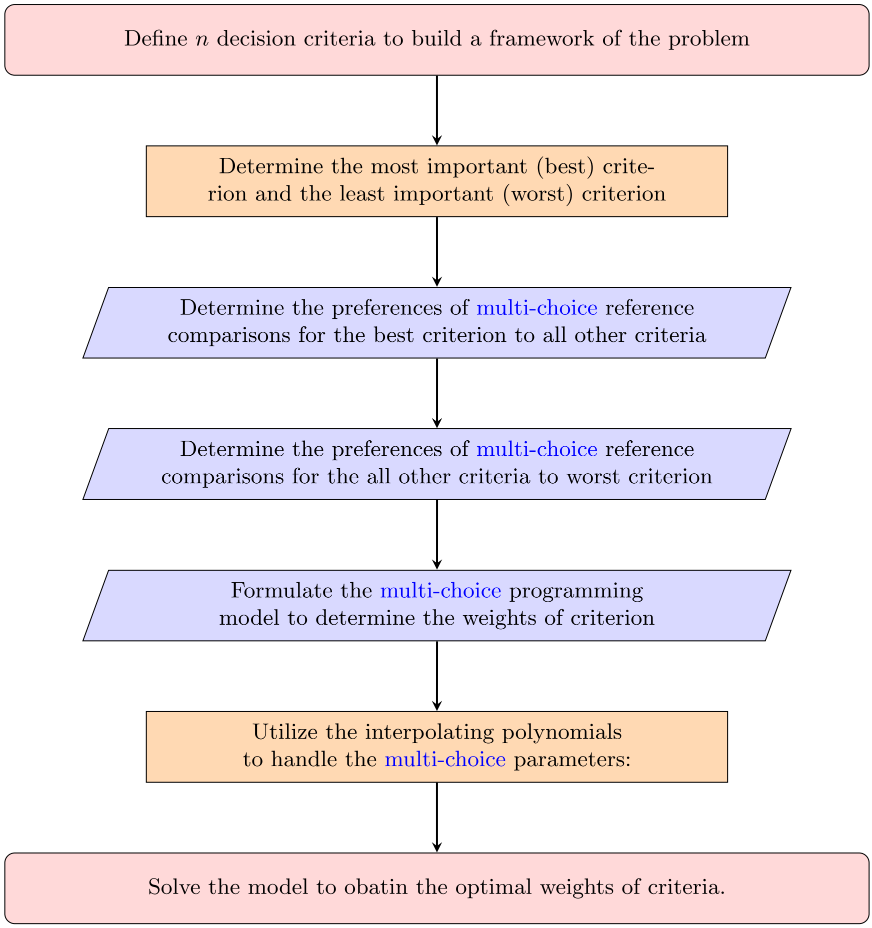

FIGURE 2 Procedure of proposed multi-choice best-worst method [Color figure can be viewed at wileyonlinelibrary.com]

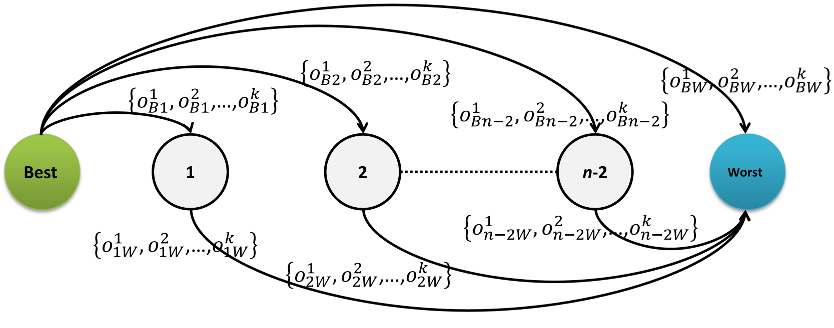

FIGURE 3 Multi-choice pairwise reference comparisons [Color figure can be viewed at wileyonlinelibrary.com]

Similarly, others-to-worst (O_W) vector presenting the pairwise comparison of preferences between all criteria to the worst criterion having a set of k number of choices is given in Equation (8) as

$$O_W = (o_{1W}, o_{2W}, o_{3W}, \ldots, o_{nW}), \quad \text{where} \quad o_{jW} = \left\{o_{jW}^1, o_{jW}^2, o_{jW}^3, \ldots, o_{jW}^k\right\} \quad \forall j, \tag{8}$$

where, o_{jW}^{k} represents the collection degree of multi-choice preference regarding the worst criterion C_W with other criterion C_j such that $o_{jW}^{k} \geq 1, j = 1, 2, ..., n$ and $k \subseteq \{1, 2, 3, ..., 9\}$.

Step 3. Find the optimal weights $(W_1^*, W_2^*, ..., W_n^*)$ to rank the criteria by using the MCPP as:

$$\textbf{Model 1:} \quad \underset{j}{\text{Min Max}} \left\{ \left| \frac{W_B}{W_j} - \left\{ o_{Bj}^1, o_{Bj}^2, ..., o_{Bj}^k \right\} \right|, \left| \frac{W_j}{W_W} - \left\{ o_{jW}^1, o_{jW}^2, ..., o_{jW}^k \right\} \right| \right\} \tag{9}$$
$$\text{subject to} \quad \sum W_j = 1, W_j \geq 0, \; j = 1, 2, ..., n.$$

The above mathematical model can be easily transformed into following model:

$$\textbf{Model 2:} \quad \text{Min} \quad \xi \tag{10}$$

$$\text{subject to} \quad \left| W_B - W_j \times \left\{ o_{Bj}^1, o_{Bj}^2, ..., o_{Bj}^{k_j} \right\} \right| \leq \xi, \forall j \tag{11}$$

$$\left| W_j - W_W \times \left\{ o_{jW}^1, o_{jW}^2, ..., o_{jW}^{k_j} \right\} \right| \leq \xi, \forall j \tag{12}$$

$$\xi \geq 0, \sum W_j = 1, W_j \geq 0, \; j = 1, 2, ..., n.$$

Step 4. Utilize the Lagrange interpolating polynomial method[79] to handle the multi-choices involved within constraints in Equation (11) and (12) and determine the best choice to minimize the inconsistency.

Let z^{Bj} and z^{jW} represents the number of node points whose values are $(0, 1, 2, ..., k_j - 1)$ with respect to o_{Bj} and o_{jW}. Thus, $\{o_{Bj}^1, o_{Bj}^2, o_{Bj}^3, ..., o_{Bj}^{k_j}\}$ and $\{o_{jW}^1, o_{jW}^2, o_{jW}^3, ..., o_{jW}^{k_j}\}$ are the associated functional values of the interpolating polynomial at k different node points. Table 2 represents the node points and its functional values involved in determining polynomials for constraints. For each multi-choice parameters in the constraint functions Equations (11) and (12), we formulate an interpolating polynomial $P_{k_j-1}(z^{Bj})$ and $P_{k_j-1}(z^{jW})$ both of degree $(k_j - 1)$ for $j - th$ criterion by using the Lagrange interpolation formula as given below.

TABLE 2 Node points and its functional values

Polynomial	Descriptions					
$P_{k_j-1}(z^{Bj})$	z^{Bj}	0	1	2	⋯	$k_j - 1$
	$f(z^{Bj})$	o_{Bj}^1	o_{Bj}^2	o_{Bj}^3	⋯	$o_{Bj}^{k_j}$
$P_{k_j-1}(z^{jW})$	z^{jW}	0	1	2	⋯	$k_j - 1$
	$f(z^{Bj})$	o_{Bj}^1	o_{Bj}^2	o_{Bj}^3	⋯	$o_{Bj}^{k_j}$

$$P_{k_j-1}(z^{Bj}) = \frac{(z^{Bj}-1)(z^{Bj}-2)\cdots(z^{Bj}-k+1)}{(-1)^{k-1}(k-1)!}o_{Bj}^{1} + \frac{z^{Bj}(z^{Bj}-2)\cdots(z^{Bj}-k+1)}{(-1)^{k-2}(k-2)!}o_{Bj}^{2} + \frac{z^{Bj}(z^{Bj}-1)(z^{Bj}-3)\cdots(z^{Bj}-k+1)}{(-1)^{k-3}(k-3)!2!}o_{Bj}^{3} + \cdots + \frac{z^{Bj}(z^{Bj}-1)(z^{Bj}-2)\cdots(z^{Bj}-k+2)}{(k-1)!}o_{Bj}^{k}\ \forall j, \tag{13}$$

$$P_{k_j-1}(z^{jW}) = \frac{(z^{jW}-1)(z^{jW}-2)\cdots(z^{jW}-k+1)}{(-1)^{k-1}(k-1)!}o_{jW}^{1} + \frac{z^{jW}(z^{jW}-2)\cdots(z^{jW}-k+1)}{(-1)^{k-2}(k-2)!}o_{jW}^{2} + \frac{z^{ijW}(z^{jW}-1)(z^{jW}-3)\cdots(z^{jW}-k+1)}{(-1)^{k-3}(k-3)!2!}o_{jW}^{3} + \cdots + \frac{z^{jW}(z^{jW}-1)(z^{jW}-2)\cdots(z^{jW}-k+2)}{(k-1)!}o_{jW}^{k}\ \forall j. \tag{14}$$

Then, the transformed optimization model for MCBWM as

Model 3 : Min ξ
subject to (15)

$$W_B - W_j \times P_{k_j-1}(z^{Bj}) \le \xi, \quad \forall j, \tag{16}$$

$$W_B - W_j \times P_{k_j-1}(z^{Bj}) \ge -\xi, \quad \forall j, \tag{17}$$

$$W_j - W_W \times P_{k_j-1}(z^{jW}) \le \xi, \quad \forall j, \tag{18}$$

$$W_j - W_W \times P_{k_j-1}(z^{jW}) \ge -\xi, \quad \forall j, \tag{19}$$

$$\sum W_j = 1, \tag{20}$$

$$W_j \ge 0, \quad \forall j, \tag{21}$$

$$\xi \ge 0, \tag{22}$$

$$z^{Bj} = 0, 1, 2, ..., (k_j - 1) \quad \forall j = 1, 2, 3, ..., m, \tag{23}$$

$$z^{jW} = 0, 1, 2, ..., (k_j - 1) \quad \forall j = 1, 2, 3, ..., m. \tag{24}$$

The MCPP as defined in Model 3 can be solved for the decision variables, that is, weight of criteria W_j, z^{Bj}, and z^{jW} for all $j = 1, 2, ..., n$.

After solving Model 3, the optimal weights and the best choice will be obtained. The reliability of pairwise comparisons has been checked using the consistency ratio (CR)[18] defined as

$$Consistency\ Ratio = \frac{\xi^*}{Consistency\ Index}, \tag{25}$$

where ξ^* is the optimal objective value of Model 3 and consistency index (CI) is a fixed value given in Table 3. The values of CI are obtained by o_{BW}, determined after solving the mathematical model. For example, if initially $o_{BW} = \{o_{BW}^1, o_{BW}^2\}$ and after solving the node point is $Z^{BW} = 1$ than CI will be considered for the value of $o_{BW} = o_{BW}^2$. Similarly, if node point

TABLE 3 CI[18]

o_{BW}	1	2	3	4	5	6	7	8	9
CI (max ξ)	0.00	0.44	1	1.63	2.30	3	3.73	4.47	5.23

TABLE 4 Multi-choice pairwise comparisons for the best criterion in Case Study-1

Criteria	C_1	C_2	C_3
Best criteria: C_3	{7, 8}	2	1

TABLE 5 Multi-choice pairwise comparisons for the worst criterion in Case Study-1

Criteria	Worst criteria: C_1
C_1	1
C_2	{5, 7}
C_3	{7, 8}

obtained is $Z^{BW} = 0$ then CI will be considered for the value of $o_{BW} = o^1_{BW}$. If CR is zero, it implies that the solution is fully consistent, and as CR increases, the consistency decreases.

4 | EXPERIMENTAL STUDIES

In this section, three real-life case studies are selected for the demonstration of the application of the proposed MCBWM approach. Further, the results are verified by comparing them with other BWM approaches.

4.1 | Case Study-1

An organization wants to choose the best possible form of transportation to ship the goods to the shop. As the Case Sudy-1 of the experimental simulation, we incorporate this same illustration of transportation management innovation discussed in [18]. Rezaei used the BWM strategy for dealing with this issue.[18] Because of the judgment maker's confusion and intangibility while experimenting, the MCBWM is used to find the best optimum means of transport.

The criteria identified to determine the optimal mode of transportation are load flexibility (C_1), accessibility (C_2), and cost (C_3) (Step-1). The flexibility (C_1) and cost (C_3) are considered as the worst and best criterion, respectively, that is, $C_B = C_3$ and $C_W = C_1$. The multi-choice pairwise comparisons for the best criterion to all other criteria are provided in Table 4 and all other criteria to the worst criterion are provided in Table 5. The best criterion C_3 has two choices in preference over the C_1 as either very important or intermediate of very and extremely important provided by the opinion of decision maker. Similarly, for the preference of C_2 criterion over the worst criterion C_1 the decision maker has choices as either important over very important. Using Tables 1, 4, and 5 presents the multi-choices best and worst parameters. Thus, $O_B = (o_{B1}, o_{B2}, o_{B3})$, that is $O_B = (o_{31}, o_{32}, o_{33})$ where $o_{31} = \{o^1_{31} = 7, o^2_{31} = 8\}, o_{32} = 2, o_{33} = 1$ as in Equation (7) and $O_W = (o_{1W}, o_{2W}, o_{3W})$, that is, $O_W = (o_{11}, o_{21}, o_{31})^T$ where $o_{11} = 1, o_{21} = \{o^1_{21} = 5, o^2_{21} = 7\}$, and $o_{31} = \{o^1_{31} = 7, o^2_{31} = 8\}$ from Equation (8) (Step-2).

Considering the details presented in Tables 4, and 5, Model-2, in the Step-3, for this problem is given as follows:

$$
\begin{aligned}
&\text{Min} \quad \xi \\
&\text{s. t.} \\
&\qquad \left| W_3 - W_j \times \left\{ o_{3j}^{k_j} \right\} \right| \leq \xi, \quad \forall j, \qquad (26) \\
&\qquad \left| W_j - W_1 \times \left\{ o_{j1}^{k_j} \right\} \right| \leq \xi, \quad \forall j, \qquad (27) \\
&\qquad \xi \geq 0 \\
&\qquad \sum W_j = 1, W_j \geq 0, \quad \forall j.
\end{aligned}
$$

To achieve consistency, Equations (26) and (27) must be established. The multiple choices involved within $o_{3j}^{k_j}$ and $o_{j1}^{k_j}$ are managed by using the Lagrange interpolating polynomials (Step-4). For example, $o_{31} = \{o_{31}^1 = 7, o_{31}^2 = 8\}$, we have the node point represented by $z^{31} \in 0, 1$ (as $k_j = 2$) and for $o_{21} = \{o_{21}^1 = 5, o_{21}^2 = 7\}$ the node point is $z^{21} \in 0, 1$. From Table 2 and using Equations (13) (14), the calculation of Lagrangian polynomials for both the node points are as follows:

1) For Two choices: $k_j = 2$, that is, $o_{Bj} = (o_{Bj}^1, o_{Bj}^2)$

$$
\begin{aligned}
P_1(z^{Bj}) &= -(z^{Bj} - 1)o_{Bj}^1 + z^{Bj} o_{Bj}^2 \\
P_1(z^{31}) &= -(z^{31} - 1)o_{31}^1 + z^{31} o_{31}^2 \\
&= -(z^{31} - 1)7 + z^{31}8 = z^{31} + 7.
\end{aligned} \tag{28}
$$

2) For Two choices: $k_j = 2$, that is, $o_{jW} = (o_{jW}^1, o_{jW}^2)$

$$
\begin{aligned}
P_1(z^{jW}) &= -(z^{jW} - 1)o_{jW}^1 + z^{jW} o_{jW}^2 \\
P_1(z^{21}) &= -(z^{21} - 1)o_{21}^1 + z^{21} o_{21}^2 \\
&= -(z^{21} - 1)5 + z^{21}7 = 2z^{21} + 5.
\end{aligned} \tag{29}
$$

Hence, we formulated a mixed integer constrained optimization problem, using model 3, as follows:

$$
\begin{aligned}
&\text{Min} \quad \xi \\
&\text{s. t.} \\
&\qquad W_3 - W_1 \times (z^{31} + 7) \leq \xi; \\
&\qquad W_3 - W_1 \times (z^{31} + 7) \geq -\xi; \\
&\qquad W_3 - W_2 \times 2 \leq \xi; \\
&\qquad W_3 - W_2 \times 2 \geq -\xi; \\
&\qquad W_2 - W_1 \times (2z^{21} + 5) \leq \xi; \\
&\qquad W_2 - W_1 \times (2z^{21} + 5) \geq -\xi; \\
&\qquad \xi \geq 0; z^{31}, z^{21} \in \{0, 1\} \\
&\qquad \sum W_j = 1; W_j \geq 0; j = 1, 2, 3.
\end{aligned} \tag{30}
$$

After solving the above model, the optimal weight values are $W_1^* = 0.0714$, $W_2^* = 0.3214$, and $W_3^* = 0.6071$ for criteria "load flexibility," "accessibility," and "cost," respectively. The optimal weights of criteria $W_1^* = 0.0714$, $W_2^* = 0.3387$, and $W_3^* = 0.5899$ are obtained through the implementation of the BWM procedure developed by Rezaei.[18] Also, weights of three criteria are $W_1^* = 0.1431$, $W_2^* = 0.3496$, and $W_3^* = 0.5073$, respectively, via the fuzzy BWM method proposed by Guo and Zhao.[58] The optimal weight of criteria and their rank orders of both of these methods along with proposed are provided in Table 6. From Table 6, it could be observed that even the ranks of three criteria are similar for BWM,[18] fuzzy BWM[58] and proposed MCBWM. However, there are slight differences seen in weights of criteria.

For consistency ratio, the optimal value of ξ^* is 0.03571 and the node points are obtained as $Z_{31} = 1$ and $Z_{21} = 0$. Therefore, the suited choice for $o_{31} = 8$ for $Z_{31} = 1$ and $o_{21} = 5$ for $Z_{21} = 0$. Since $o_{BW} = o_{31} = 8$, the value of consistency index from Table 3 for this case study is 4.47. The consistency ratio $= 0.03571/4.47 = 0.00799$, that implies the solution values are highly consistent as it very near to zero. The consistency ratio in [58] for this problem by solving fuzzy BWM is 0.0559 whereas in [18], we found the consistency ratio for the same case study by solving BWM is 0.058. The consistency ratio for proposed MCBWM approach is much-much better in comparison to these two. A pictorial comparison of consistency ratios of these three approaches for this problem is shown in Figure 4. From Figure 4, it is clearly seen that the MCBWM shows much lesser inconsistency that the fuzzy BWM and BWM methods.

4.1.1 | Analysis of Case Study-1

An analysis has been done to see change in weights, consistency and optimal set of responses obtained on solving the problem for multiple combination of choices. We have considered the

TABLE 6 Ranking and optimal weights of criteria for Case Study-1

Methods	W_1	W_2	W_3	Ordering
Rezaei[18]	0.0714	0.3387	0.5899	$C_3 \geq C_2 \geq C_1$
Guo & Zhao[58]	0.1431	0.3496	0.5073	$C_3 \geq C_2 \geq C_1$
Proposed	0.0714	0.3214	0.6071	$C_3 \geq C_2 \geq C_1$

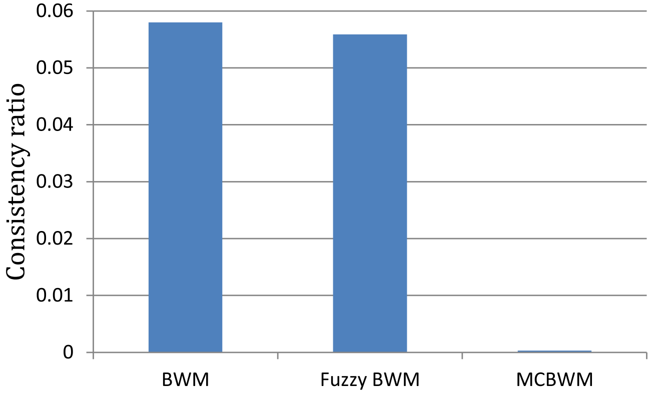

FIGURE 4 Comparison of consistency ratios for Case Study-1 [Color figure can be viewed at wileyonlinelibrary.com]

same data used in Case Study-1 and considered different values of choices for $o_{21} \in \{1, 2, 3, ..., 9\}$. A total of 64 possible combinations of choices are evaluated. Table 7 shows ξ^*, w_1, w_2, w_3, Z^{21} and Z^{31} for all possible combinations of choices of o_{21}, that is, $\{o_{21}^1, o_{21}^2\}$. From Table 7, It is clear that changing the choice values puts a relative change in ξ^* and weights. Also, the optimal values of weights are obtained by choosing the best choice among choices of preferences, that is, o_{21} and o_{31}. Full consistency, that is, $\xi^* = 0$ is achieved when $o_{32} \times o_{21} = o_{31}$. It is nothing but $o_{Bj} \times o_{jW} = o_{BW}$. Here, o_{32} is 2 and o_{21} is always chosen as 4 out of all pair of choices, so that the product value will become equals to $o_{31} = 8$. The maximum value of ξ^* obtained is 0.1389. The consistency ratio for all possible choices of o_{21}^1 and o_{21}^2 are presented in Table 8 and Figure 5. The figure is plotted considering the o_{21}^1 on x-axis. It is clearly visible that for $o_{21}^1 = 4$ and $o_{21}^2 = 4$ the CR value is zero. It has been found that the CR values obtained are very much less than the CR values for BWM[18] and Fuzzy BWM[58] approach.

4.2 | Case Study-2:

In this case study, the goal is to select a high performance car.[19] The decision is taken to buy a car from a set of alternatives that are based on five criteria, namely quality (C_1), price (C_2), comfort (C_3), safety (C_4), and style (C_5). Here, the MCBWM is applied to select the best car to purchase.

Initially, for best car selection, the aforementioned five criteria are selected, and the criterion "price" and "style" are considered as the best criterion and the worst criterion, which means $C_B = C_2$ and $C_W = C_5$ (Step-1). The multi-choice pairwise comparisons for the best criterion to all other criteria are provided in Table 9 and all other criteria to the worst criterion are provided in Table 10. The best criterion C_2 has two choices in preference over the C_4 as either intermediate of equally and moderate or moderate important provided by the opinion of the decision-maker. Similarly, for the preference of the C_3 criterion over the worst criterion C_5, the decision-maker has choices as either intermediate of equally and moderate important or intermediate of moderately important and important. Also, for the preference of the C_4 criterion over the worst criterion C_5, the decision-maker has three choices as either intermediate of equally and moderate or moderate important or intermediate of moderate and important. The linguistic terms and their numeric form of buyer's preferences for best criterion to other criteria and other criteria to worst criterion are presented in Table 1.

Using Table 1, Tables 9, and 10 present the multi-choice best and worst parameters. Thus, $O_B = (o_{21}, o_{22}, o_{23}, o_{24}, o_{25})$, where $o_{21} = 2, o_{22} = 1, o_{23} = 4, o_{24} = \{o_{24}^1 = 2, o_{24}^2 = 3\}$, and $o_{25} = 8$, as in Equation (7). And, $O_W = (o_{15}, o_{25}, o_{35}, o_{45}, o_{55})^T$ where $o_{15} = 4, o_{25} = 8, o_{35} = \{o_{35}^1 = 2, o_{35}^2 = 4\}, o_{45} = \{o_{45}^1 = 2, o_{45}^2 = 3, o_{45}^3 = 4\}$ and $o_{55} = 1$ from Equation (8) (Step-2). The Model-2 for this problem is given as follows (Step-3):

$$\begin{aligned} &\text{Min} \quad \xi \\ &\text{subject to} \end{aligned} \tag{31}$$

$$\left| W_2 - W_j \times \left\{ o_{2j}^{k_j} \right\} \right| \leq \xi \quad \forall j, \tag{32}$$

$$\left| W_j - W_5 \times \left\{ o_{j5}^{k_j} \right\} \right| \leq \xi \quad \forall j, \tag{33}$$

$$\sum_{j=1}^{5} W_j = 1, W_j \geq 0, \forall j = 1, 2, ..., 5, \xi \geq 0.$$

TABLE 7 Solution values of Case Study-1 for possible choice sets of $o_{21} = \{o_{21}^1, o_{21}^2\}$

o_{21}^1		o_{21}^2 1	2	3	4	5	6	7	8
1	ξ^*	0.1389	0.0750	0.0227	0.0000	0.0357	0.0666	0.0938	0.1176
	w_1	0.1111	0.1000	0.0909	0.0769	0.0714	0.0666	0.0625	0.0588
	w_2	0.2500	0.2750	0.2955	0.3077	0.3214	0.3333	0.3438	0.3529
	w_3	0.6389	0.6250	0.6136	0.6154	0.6071	0.6000	0.5938	0.5882
	Z^{21}	0	1	1	1	1	1	1	1
	Z^{31}	0	0	0	1	1	1	1	1
2	ξ^*	0.0750	0.0750	0.0227	0.0000	0.0357	0.0667	0.0750	0.0750
	w_1	0.1000	0.1000	0.0909	0.0769	0.0714	0.0667	0.1000	0.1000
	w_2	0.2750	0.2750	0.2955	0.3077	0.3214	0.3333	0.2750	0.2750
	w_3	0.6250	0.6250	0.6136	0.6154	0.6071	0.6000	0.6250	0.6250
	Z^{21}	0	0	1	1	1	1	0	0
	Z^{31}	0	0	0	1	1	1	0	0
3	ξ^*	0.0227	0.0227	0.0227	0.0000	0.0227	0.0227	0.0227	0.0227
	w_1	0.0909	0.0909	0.0909	0.0769	0.0909	0.0909	0.0909	0.0909
	w_2	0.2955	0.2955	0.2955	0.3077	0.2955	0.2955	0.2955	0.2955
	w_3	0.6136	0.6136	0.6136	0.6154	0.6136	0.6136	0.6136	0.6136
	Z^{21}	0	0	0	1	0	0	0	0
	Z^{31}	0	0	0	1	0	0	0	0
4	ξ^*	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	w_1	0.0769	0.0769	0.0769	0.0769	0.0769	0.0769	0.0769	0.0769
	w_2	0.3077	0.3077	0.3077	0.3077	0.3077	0.3077	0.3077	0.3077
	w_3	0.6154	0.6154	0.6154	0.6154	0.6154	0.6154	0.6154	0.6154
	Z^{21}	0	0	0	0	0	0	0	0
	Z^{31}	1	1	1	1	1	1	1	1
5	ξ^*	0.0357	0.0357	0.0227	0.0000	0.0357	0.0357	0.0357	0.0357
	w_1	0.0714	0.0714	0.0909	0.0769	0.0714	0.0714	0.0714	0.0714
	w_2	0.3214	0.3214	0.2955	0.3077	0.3214	0.3214	0.3214	0.3214
	w_3	0.6071	0.6071	0.6136	0.6154	0.6071	0.6071	0.6071	0.6071
	Z^{21}	0	0	1	1	0	0	0	0
	Z^{31}	1	1	0	1	1	1	1	1
6	ξ^*	0.0667	0.0667	0.0227	0.0000	0.0357	0.0667	0.0667	0.0667
	w_1	0.0667	0.0667	0.0909	0.0769	0.0714	0.0667	0.0667	0.0667
	w_2	0.3333	0.3333	0.2955	0.3077	0.3214	0.3333	0.3333	0.3333

(Continues)

TABLE 7 (Continued)

o_{21}^1		o_{21}^2 1	2	3	4	5	6	7	8
	w_3	0.6000	0.6000	0.6136	0.6154	0.6071	0.6000	0.6000	0.6000
	Z^{21}	0	0	1	1	1	0	0	0
	Z^{31}	1	1	0	1	1	1	1	1
7	ξ^*	0.0938	0.0750	0.0227	0.0000	0.0357	0.0667	0.0938	0.0938
	w_1	0.0625	0.1000	0.0909	0.0769	0.0714	0.0667	0.0625	0.0625
	w_2	0.3438	0.2750	0.2955	0.3077	0.3214	0.3333	0.3438	0.3438
	w_3	0.5938	0.6250	0.6250	0.6154	0.6071	0.6000	0.5938	0.5938
	Z^{21}	0	1	1	1	1	1	0	0
	Z^{31}	1	0	0	1	1	1	1	1
8	ξ^*	0.1176	0.0750	0.0227	0.0000	0.0357	0.0667	0.0938	0.1176
	w_1	0.0588	0.1000	0.0909	0.0769	0.0714	0.0667	0.0625	0.0588
	w_2	0.3529	0.2750	0.2955	0.3077	0.3214	0.3333	0.3438	0.3529
	w_3	0.5882	0.6250	0.6136	0.6154	0.6071	0.6000	0.5938	0.5882
	Z^{21}	0	1	1	1	1	1	1	0
	Z^{31}	1	0	0	1	1	1	1	1

TABLE 8 Consistency ratio of Case Study-1 for possible choice sets of $o_{21} = \{o_{21}^1, o_{21}^2\}$

CR o_{21}^1	o_{21}^2 1	2	3	4	5	6	7	8
1	0.0372	0.0201	0.0061	0.0000	0.0080	0.0149	0.0210	0.0263
2	0.0201	0.0201	0.0061	0.0000	0.0080	0.0149	0.0201	0.0201
3	0.0061	0.0061	0.0061	0.0000	0.0061	0.0061	0.0061	0.0061
4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5	0.0080	0.0080	0.0061	0.0000	0.0080	0.0080	0.0080	0.0080
6	0.0149	0.0149	0.0061	0.0000	0.0080	0.0149	0.0149	0.0149
7	0.0210	0.0201	0.0061	0.0000	0.0080	0.0149	0.0210	0.0210
8	0.0263	0.0201	0.0061	0.0000	0.0080	0.0149	0.0210	0.0263

The multi-choices involved within $o_{2j}^{k_j}$ and $o_{j5}^{k_j}$ are managed by using the Lagrange interpolating polynomials (Step-4). From Table 2 and using Equation (13) and Equation (14), the calculation of Lagrangian polynomials are as follows:

1) For $o_{24} = \{o_{24}^1 = 2, o_{24}^2 = 3\}$, we have $k_j = 2$ and node points $z^{24} \in 0, 1$, thus

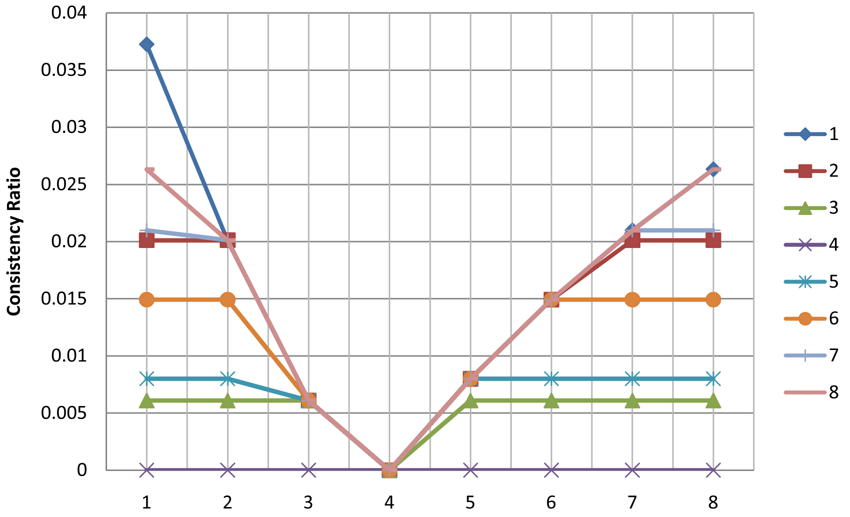

FIGURE 5 Consistency ratio of Case Study-1 for possible choice sets of $o_{21} = \left\{o_{21}^1, o_{21}^2\right\}$ [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 9 Multi-choice pairwise comparisons for the best criterion in Case Study-2

Criteria	C_1	C_2	C_3	C_4	C_5
Best criteria: C_2	2	1	4	{2, 3}	8

TABLE 10 Multi-choice pairwise comparisons for the worst criterion in Case Study-2

Criteria	Worst criteria: C_5
C_1	4
C_2	8
C_3	{2, 4}
C_4	{2, 3, 4}
C_5	1

$$\begin{aligned} P_1(z^{Bj}) &= -(z^{Bj} - 1)o_{Bj}^1 + z^{Bj}o_{Bj}^2 \\ P_1(z^{24}) &= -(z^{24} - 1)o_{24}^1 + z^{24}o_{24}^2 \\ &= -(z^{24} - 1)2 + z^{24}3 = z^{24} + 2. \end{aligned} \tag{34}$$

Similarly, for $o_{35} = \{o_{35}^1 = 2, o_{35}^2 = 4\}$, we have $k_j = 2$ and node points $z^{35} \in 0, 1$ thus $P_1(z^{35}) = 2z^{35} + 2$.

2) For three choices: $k_j = 3$, $o_{45} = \{o_{45}^1 = 2, o_{45}^2 = 3, o_{45}^3 = 4\}$ and node points $z^{45} \in 0, 1, 2$, the polynomial is given as follows:

$$
\begin{aligned}
P_2(z^{jW}) &= \frac{(z^{jW}-1)(z^{jW}-2)}{2} o_{jW}^1 - z^{jW}(z^{jW}-2) o_{jW}^2 + \frac{z^{jW}(z^{jW}-1)}{2} o_{jW}^3 \\
P_2(z^{45}) &= \frac{(z^{45}-1)(z^{45}-2)}{2} o_{45}^1 - z^{45}(z^{45}-2) o_{45}^2 + \frac{z^{45}(z^{45}-1)}{2} o_{45}^3 \\
&= \frac{(z^{45}-1)(z^{45}-2)}{2} 2 - z^{45}(z^{45}-2) 3 + \frac{z^{45}(z^{45}-1)}{2} 4 \\
&= z^{45} + 2.
\end{aligned}
$$

Now formulating the problem using model 3, we will get a optimization problem model as

$$
\begin{aligned}
&\text{Min} \quad \xi \\
&\text{subject to} \\
&W_2 - W_1 \times 2 \le \xi, W_2 - W_1 \times 2 \ge -\xi \\
&W_2 - W_3 \times 4 \le \xi, W_2 - W_3 \times 4 \ge -\xi \\
&W_2 - W_4 \times (2 + z^{24}) \le \xi, W_2 - W_4 \times (2 + z^{24}) \ge -\xi \\
&W_2 - W_5 \times 8 \le \xi, W_2 - W_5 \times 8 \ge -\xi \\
&W_1 - W_5 \times 4 \le \xi, W_1 - W_5 \times 4 \ge -\xi \\
&W_2 - W_5 \times 8 \le \xi, W_2 - W_5 \times 8 \ge -\xi \\
&W_3 - W_5 \times (2 + 2z^{35}) \le \xi, W_3 - W_5 \times (2 + 2z^{35}) \ge -\xi \\
&W_4 - W_5 \times (2 + z^{45}) \le \xi, W_4 - W_5 \times (2 + z^{45}) \ge -\xi \\
&\sum_{j=1}^{5} W_j = 1, W_j \ge 0, \forall j = 1, 2 \ ..., \ 5, \xi \ge 0 \\
&z^{24}, z^{35} \in \{0, 1\}, z^{45} \in \{0, 1, 2\}.
\end{aligned}
\tag{35}
$$

After solving the above model, the optimal weight values are $W_1^* = 0.2105$, $W_2^* = 0.4210$, $W_3^* = 0.1053$, $W_4^* = 0.2105$ and $W_5^* = 0.0526$ for criteria quality (C_1), price (C_2), comfort (C_3), safety (C_4), and style (C_5), respectively. Weights of five criteria are $W_1^* = 0.2470$, $W_2^* = 0.2842$, $W_3^* = 0.2189$, $W_4^* = 0.1608$, and $W_5^* = 0.0891$ via the fuzzy BWM method.[58] The optimal weights of criteria $W_1^* = 0.2105$, $W_2^* = 0.4211$, $W_3^* = 0.1053$, $W_4^* = 0.2105$, and $W_5^* = 0.0526$ are obtained for same problem by the BWM.[19] The optimal weight of criteria and their rank orders of both these method along with proposed are provided in Table 11. From Table 11, it could be observed that the ordering of five criteria are similar for BWM,[19] fuzzy BWM[58] and proposed MCBWM. Therefore, the ranking order obtained is *price > quality > safety > comfort > style*.

For consistency ratio, the optimal value of $\xi^*=0$ and the node points are obtained as $Z_{24} = 0$, that is, $o_{24} = 2$, $Z_{35} = 0$, that is, $o_{35} = 2$ and $Z_{45} = 2$, that is, $o_{45} = 3$. As $\xi^* = 0$, the consistency ratio is 0, which implies that the solution values are fully consistent. The consistency ratio in [19] for this problem by solving BWM is 0 whereas in [58], the

TABLE 11 Ranking and optimal weights of criteria for Case Study-2

Methods	W_1	W_2	W_3	W_4	W_5	Ordering
Rezaei[19]	0.2105	0.4211	0.1053	0.2105	0.0526	$C_2 \ge C_1 \ge C_4 \ge C_3 \ge C_5$
Guo & Zhao[58]	0.2470	0.2842	0.2189	0.1608	0.0891	$C_2 \ge C_1 \ge C_4 \ge C_3 \ge C_5$
Proposed	0.2105	0.4211	0.1053	0.2105	0.0526	$C_2 \ge C_1 \ge C_4 \ge C_3 \ge C_5$

consistency ratio for the same case study by solving using fuzzy BWM is 0.0984. Hence, the consistency ratio for proposed MCBWM approach is better in comparison with these BWM and fuzzy BWM.

In [19], the problem has been solved separately for each set of pairwise comparisons. A total of three cases are presented. Whereas, for MCBWM, the three different problem data have been assumed in a single step procedure by considering in the form of multiple choices. The presence of multiple choices involved within the parameter value provides the advantage of flexibility in the judgment of decision-makers. Moreover, it can be concluded that the MCBWM shows optimal solution using the multiple choices of the comparisons provided by the decision-makers along with the same ordering of priorities of criteria as obtained by BWM and fuzzy BWM.

4.3 | Case Study-3

In this case study, we have considered a problem of cloud service selection from e-commerce and cloud service industry, which was earlier studied by Hussain et al.[64] The problem evaluates services from the Quality of Service (QoS) and Quality of Experience (QoE) perspective. We have simulated the problem for the MCBWM approach and compared the solution with the BWM approach solution. The goal is to rank the QoS criteria (QoSc) and QoE criteria (QoEc). The criteria are identified according to the needs of the organization. The criteria sets are presented in Table 12. The pairwise reference comparisons are presented and tabulated in Tables 13 and 14.

TABLE 12 QoSc and QoEc sets for service selection

QoSc	Notation	QoEc	Notation
CPU speed	T1	Security	N1
Core concurrency	T2	Response time	N2
Memory bandwidth	T3	Accessibility	N3
Disk rate	T4	Features	N4
Disk seeks	T5	Ease of use	N5
Latency	T6	Technical support	N6
Availability	T7	Customer service	N7
Price	T8	Trust	N8

TABLE 13 Multi-choice pairwise comparisons for the best QoSc and QoEc in Case Study-3

QoSc	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
Best QoSc: T_1	1	{3, 5}	2	{2, 4}	4	{5, 7, 9}	2	9
QoEc	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_8
Best QoEc: N_1	1	3	{5, 7}	2	9	4	{3, 5, 7}	{4, 6}

TABLE 14 Multi-choice pairwise comparisons for the worst QoSc and QoEc in Case Study-3

QoSc	Worst QoSc: T_8	QoEc	Worst QoEc: N_5
T_1	9	N_1	9
T_2	{4,6}	N_2	5
T_3	7	N_3	{3,7}
T_4	5	N_4	4
T_5	3	N_5	1
T_6	{2,3}	N_6	5
T_7	{3,5,7}	N_7	{2,4}
T_8	1	N_8	5

Initially, model-2 of MCBWM has been formulated for QoSc and QoEc criteria. The multiple choices involved are formulated by using the Lagrange interpolating polynomials (Step-4). From Table 2 and using Equations (13) and (14), the calculation of Lagrangian polynomials are incorporated in the mathematical programming model of QoSc and QoEC. The Mathematical programming model-3 for QoSc by incorporating Lagrangian polynomials for multiple choices is formulated as follows:

$$\begin{aligned}
&\text{Min} \quad \xi \\
&\text{subject to} \\
&W_1 - W_2 \times (2z^{12} + 3) \leq \xi, W_1 - W_2 \times (2z^{12} + 3) \geq -\xi \\
&W_1 - W_3 \times 2 \leq \xi, W_1 - W_3 \times 2 \geq -\xi \\
&W_1 - W_4 \times (2z^{14} + 2) \leq \xi, W_1 - W_4 \times (2z^{14} + 2) \geq -\xi \\
&W_1 - W_5 \times 4 \leq \xi, W_1 - W_5 \times 4 \geq -\xi \\
&W_1 - W_6 \times (2z^{16} + 5) \leq \xi, W_1 - W_6 \times (2z^{16} + 5) \geq -\xi \\
&W_1 - W_7 \times 2 \leq \xi, W_1 - W_7 \times 2 \geq -\xi \\
&W_1 - W_8 \times 9 \leq \xi, W_1 - W_8 \times 9 \geq -\xi \\
&W_2 - W_8 \times (2z^{28} + 4) \leq \xi, W_2 - W_8 \times (2z^{28} + 4) \geq -\xi \\
&W_3 - W_8 \times 7 \leq \xi, W_3 - W_8 \times 7 \geq -\xi \\
&W_4 - W_8 \times 5 \leq \xi, W_4 - W_8 \times 5 \geq -\xi \\
&W_5 - W_8 \times 3 \leq \xi, W_5 - W_8 \times 3 \geq -\xi \\
&W_6 - W_8 \times (z^{68} + 2) \leq \xi, W_6 - W_8 \times (z^{68} + 2) \geq -\xi \\
&W_7 - W_8 \times (2z^{78} + 3) \leq \xi, W_7 - W_8 \times (2z^{78} + 3) \geq -\xi \\
&\sum_{j=1}^{8} W_j = 1, W_j \geq 0, \forall j = 1, 2 \ ..., \ 8, \xi \geq 0 \\
&z^{12}, z^{14}, z^{28}, z^{68} \in \{0, 1\}, z^{16}, z^{78} \in \{0, 1, 2\}.
\end{aligned} \tag{36}$$

The Mathematical programming model for QoEc becomes:

$$
\begin{aligned}
&\text{Min} \quad \xi \\
&\text{subject to} \\
&W_1 - W_2 \times 3 \leq \xi, W_1 - W_2 \times 3 \geq -\xi \\
&W_1 - W_3 \times (2z^{13} + 5) \leq \xi, W_1 - W_3 \times (2z^{13} + 5) \geq -\xi \\
&W_1 - W_4 \times 2 \leq \xi, W_1 - W_4 \times 2 \geq -\xi \\
&W_1 - W_5 \times 9 \leq \xi, W_1 - W_5 \times 9 \geq -\xi \\
&W_1 - W_6 \times 4 \leq \xi, W_1 - W_6 \times 4 \geq -\xi \\
&W_1 - W_7 \times (2z^{17} + 3) \leq \xi, W_1 - W_7 \times (2z^{17} + 3) \geq -\xi \\
&W_1 - W_8 \times (2z^{18} + 4) \leq \xi, W_1 - W_8 \times (2z^{18} + 4) \geq -\xi \\
&W_2 - W_5 \times 5 \leq \xi, W_2 - W_5 \times 5 \geq -\xi \\
&W_3 - W_5 \times (4z^{35} + 3) \leq \xi, W_3 - W_5 \times (4z^{35} + 3) \geq -\xi \\
&W_4 - W_5 \times 4 \leq \xi, W_4 - W_5 \times 4 \geq -\xi \\
&W_6 - W_5 \times 5 \leq \xi, W_6 - W_5 \times 5 \geq -\xi \\
&W_7 - W_5 \times (2z^{75} + 2) \leq \xi, W_7 - W_5 \times (2z^{75} + 2) \geq -\xi \\
&W_8 - W_5 \times 5 \leq \xi, W_7 - W_8 \times 5 \geq -\xi \\
&\sum_{j=1}^{8} W_j = 1, W_j \geq 0, \forall j = 1, 2 \ldots, 8, \xi \geq 0 \\
&z^{13}, z^{18}, z^{35}, z^{37} \in \{0, 1\}, z^{17} \in \{0, 1, 2\}.
\end{aligned}
\tag{37}
$$

After solving the mathematical programming models for QoSc and QoEc criteria separately, we have got the optimal solution in terms of weight values of QoSc and QoEc criteria. To evaluate the performance of the proposed MCBWM, the case study has also been solved using BWM[19] approach.

The optimal weight values are presented in Tables 15 and 16. Also, the inconsistency and priority order of criteria obtained is presented in Table 17. It has been found that for QoSc, there is no change in weight values, inconsistency, and order of criteria using both approaches. But, In Table 16 for QoEc, their is a slight difference of weight values and inconsistency value. The inconsistency values (ξ^*) of services QoSC and QoEc are also compared and presented through Figure 6. The order of criteria obtained remains the same for both approaches. The inconsistency value for both QoSc and QoEc has been found acceptable. The results show that the MCBWM approach performs well and provided optimal results while considering multiple choices of preference comparisons.

TABLE 15 Optimal weights of QoSc criteria for Case Study-3

Methods	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
Rezaei[19]	0.2733	0.1022	0.1533	0.1533	0.0766	0.0613	0.1533	0.0267
Proposed	0.2733	0.1022	0.1533	0.1533	0.0766	0.0613	0.1533	0.0267

TABLE 16 Optimal weights of QoEc criteria for Case Study-3

Methods	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_8
Rezaei[19]	0.3132	0.1221	0.0732	0.1686	0.0289	0.0916	0.1108	0.0916
Proposed	0.3097	0.1207	0.0724	0.1668	0.0286	0.0905	0.1207	0.0905

TABLE 17 Inconsistency and ordering of QoSc and QoEc criteria in Case Study-3

Method	Service	ξ^*	Ordering
Rezaei[19]	QoSc	0.0333	$T_1 > T_3 \geq T_4 \geq T_7 > T_2 > T_5 > T_6 > T_8$
Proposed	QoSc	0.0333	$T_1 > T_3 \geq T_4 \geq T_7 > T_2 > T_5 > T_6 > T_8$
Rezaei[19]	QoEc	0.0530	$N_1 > N_4 > N_2 > N_7 > N_6 \geq N_8 > N_3 > N_5$
Proposed	QoEc	0.0524	$N_1 > N_4 > N_2 > N_7 > N_6 \geq N_8 > N_3 > N_5$

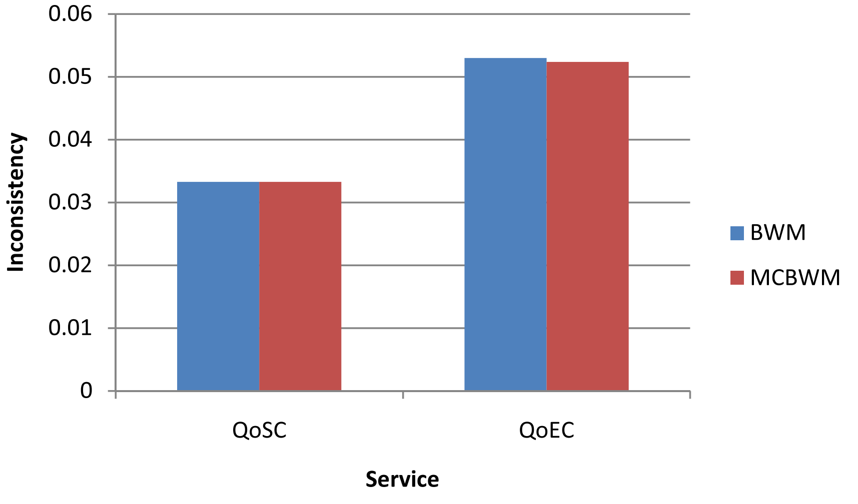

FIGURE 6 Inconsistency of QoSc and QoEc service for Case Study-3 [Color figure can be viewed at wileyonlinelibrary.com]

5 | CONCLUSION AND FUTURE WORK

In this paper, we propose a novel MCBWM, which is an extension of the BWM. The proposed MCBWM can overcome the limitation of the BWM approach. It can solve MCDMs where the pairwise reference in the comparisons is defined as multi-choice parameters and a single value of parameter with the integration of the multi-choice programming approch. To demonstrate the applicability of the proposed MCBWM, we have utilized three case studies representing real-life decision-making problems. For performance evaluation of MCBWM, case studies' results are compared with fuzzy-BWM and BWM approaches. It has been found that MCBWM is a robust method with better performance. The following features make MCBWM vital and exciting in comparison to BWM.

1. The MCBWM shows flexibility by providing the freedom to an expert to choose the multiple pairwise comparison values of criteria/alternatives. Whereas, BWM doesn't have such a feature. This gives an advantage to MCBWM on BWM concerning human perception and cognition.
2. The BWM can't solve MCDM problems if the relative pairwise reference in the comparisons is defined as multi-choice parameters, but MCBWM can solve such problems.
3. MCBWM handles multiple BWM models in a single model by using a multi-choice programming approch. It contributes to less calculation and analysis time.

4. In MCBWM, suppose the expert provides two choices for one pairwise comparison and three choices for the other, then it will lead to a total of six BWM models. However, the proposed approach manages all six BWM models in a single model. In the end, the MCBWM model provides the best optimal solution out of all choices, choosing one choice for the optimal solution.
5. In MCBWM, the CR is either less or equal to the CR of BWM which makes MCBWM more or equally reliable to the BWM.

Like other methods, this method also has some limitations. One of the limitations of this method is that it seems mathematical, which makes the method complicated for managers and some decision-makers. The formulation of multi-choice parameters to a polynomial and then computer programming to solve the mathematical programming model needs some level of technical assistance. More choices in pairwise comparisons make the modeling more complicated. In the future, these limitations may overcome by a user-oriented programmed solver specially designed for MCBWM. Such a solver will reduce the computational effort of the decision-maker.

We are working on some more real-life problems to gain insight into the proposed MCBWM for future extensions. We strongly believe that our proposed method is a novel one and will provide directions to future research. Like BWM, this method can also be developed using different data sets such as fuzzy, probabilistic, and so forth. We suggest extending the technique in a group decision-making scenario. The proposed approach can be extended to multi-criteria Group decision-making[81] and Large scale Group decision-making problems.[82] Future research with new real-life problems may provide more depth into the validity, applicability, and usefulness of the proposed method.

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